

Technology Stack

Date	13 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 (Flask templating engine)	Server-rendered templates (index.html, result.html) are sufficient for a simple form-based prediction UI, avoiding the overhead of a separate JS framework for this scope.

2	Backend / Server-Side	Python 3, Flask	Flask is lightweight and integrates directly with scikit-learn and joblib for serving the trained ML model behind a simple REST-style /predict endpoint.
3	Database / Data Storage	CSV files (training); relational schema planned (MySQL/PostgreSQL, see ER diagram)	CSV files (application_record.csv, credit_record.csv) were used for model training. The ER diagram defines a relational schema (Users, Applicant_Details, Credit_History, ML_Model, Approval_Prediction) for persisting user accounts and prediction history in a future/extended version of the app.
4	Cloud / Hosting / Deployment	Render / Heroku (Gunicorn WSGI server)	Gunicorn (listed in requirements.txt) is the standard production WSGI server for deploying Flask apps on platforms like Render or Heroku, enabling easy, low-cost hosting for a small ML web app.
5	Version Control & CI/CD	Git & GitHub	Git/GitHub is used for version control and collaboration on the codebase and notebook. No automated CI/CD pipeline is currently implemented.

6	Third-Party APIs / Other Tools	scikit-learn, pandas, NumPy, joblib	scikit-learn provides the RandomForestClassifier used for approval prediction; pandas/NumPy handle data preprocessing; joblib serializes and loads the trained model (credit_card_model.pkl) for use in the Flask app.